

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTMC502 - MACHINE DESIGN - I

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 What are the two types of rigid flange couplings in machine design?
 - a) Stable and unstable
 - b) Flexible and non-flexible
 - c) Protected and unprotected
 - d) Deformable and non-deformable
- 2 Which of the following materials are subjected to most severe stress concentration?
 - a) Ductile materials under fluctuating load
 - b) Brittle materials under static and fluctuating load
 - c) Brittle materials under fluctuating load
 - d) Ductile materials under static load
- 3 Which type of failures does modified goodman line combines?
 - a) Fatigue and yield
 - b) Bending and fatigue
 - c) Yield and bending
 - d) Fatigue, bending and yield
- 4 Which theory of failure is mostly used for ductile materials in machine design?
 - a) Maximum principal stress theory
 - b) Maximum shear stress theory
 - c) Distortion energy theory
 - d) Haigh's theory
- 5 Which of the following is not a type of welding joint?
 - a) Lap joint
 - b) Butt joint
 - c) Tee joint
 - d) Single strap joint
- 6 Which of the following shaft are the integrals or equally spaced lines made on a shaft?
 - a) Stepped shaft
 - b) Spline shaft
 - c) Cam shaft
 - d) Crankshaft
- 7 Which of the following is a dominance of power screw?
 - a) Large load carrying capacity
 - b) High efficiency
 - c) Wear of screw is reduced
 - d) Mechanical advantage is lowered

- 8 Which of the following function can the spring perform?
 - a) Store energy
 - b) Absorb shock
 - c) Measure force
 - d) All of the mentioned
- 9 Ergonomics in design deals with:
 - a) The aesthetic appeal of a product
 - b) Material selection for mechanical strength
 - c) Human interaction and comfort in using the product
 - d) Fatigue resistance of the material
- 10 Which of the following describes fatigue failure?
 - a) Sudden failure under cyclic loading
 - b) Slow deformation under a constant load
 - c) Instantaneous fracture due to impact
 - d) Gradual wear and tear of material
- 11 Which design phase focuses on understanding how the forces affect the components?
 - a) Material selection
 - b) Force analysis
 - c) Ergonomic design
 - d) Economic consideration
- 12 What is the primary consideration when designing a shaft subjected to a bending moment?
 - a) Shear Stress
 - b) Tensile Stress
 - c) Bending Stress
 - d) Compressive Stress

Q2. Solve the following.

- A) What are the steps involved in design of a machine element? 6
- B) Explain man-machine joint system. 6

Q3. Solve the following.

- A) The stresses induced at a critical point in a machine component made of steel 45C8 ($S_{yt} = 380 \text{ N/mm}^2$) are 6
as follows: $\sigma_x = 100 \text{ N/mm}^2$ $\sigma_y = 40 \text{ N/mm}^2$ $\tau_{xy} = 80 \text{ N/mm}^2$ Calculate the factor of safety by
(i) the maximum normal stress theory,
(ii) the maximum shear stress theory, and
(iii) the distortion energy theory.
- B) Two rods, made of plain carbon steel 40C8 ($S_{yt} = 380 \text{ N/mm}^2$), are to be connected by means of a cotter 6
joint. The diameter of each rod is 50 mm and the cotter is made from a steel plate of 15 mm thickness.
Calculate the dimensions of the socket end making the following assumptions:
(i) the yield strength in compression is twice of the tensile yield strength; and
(ii) the yield strength in shear is 50% of the tensile yield strength.
The factor of safety is 6.

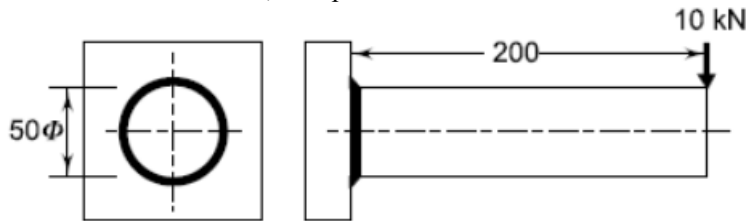
Q4. Solve Any Two of the following.

- A) Explain the Gerber curve and Soderberg and Goodman lines? 6
- B) A machine component is subjected to fluctuating stress that varies from 40 to 100 N/mm^2 . The corrected 6
endurance limit stress for the machine component is 270 N/mm^2 . The ultimate tensile strength and yield
strength of the material are 600 and 450 N/mm^2 respectively. Find the factor of safety using
(i) Gerber theory
(ii) Soderberg line
(iii) Goodman line
Also, find the factor of safety against static failure.

- C) A rotating shaft, 40 mm in diameter, is made of steel FeE 580 ($S_{yt} = 580 \text{ N/mm}^2$). It is subjected to a steady torsional moment of 250 N-m and bending moment of 1250 N-m. Calculate the factor of safety based on, 6
- maximum principal stress theory; and
 - maximum shear stress theory.

Q5. Solve Any Two of the following.

- A) A rigid coupling is used to transmit 50 kW power at 300 rpm. There are six bolts. The outer diameter of the flanges is 200 mm, while the recess diameter is 150 mm. The coefficient of friction between the flanges is 0.15. The bolts are made of steel 45C8 ($S_{yt} = 380 \text{ N/mm}^2$) and the factor of safety is 3. Determine the diameter of the bolts. Assume that the bolts are fitted in large clearance holes. 6
- B) The nominal diameter of a triple threaded square screw is 50 mm, while the pitch is 8 mm. It is used with a collar having an outer diameter of 100 mm and inner diameter as 65 mm. The coefficient of friction at the thread surface as well as at the collar surface can be taken as 0.15. The screw is used to raise a load of 15 kN. Using the uniform wear theory for collar friction, calculate: 6
- torque required to raise the load;
 - torque required to lower the load; and
 - the force required to raise the load, if applied at a radius of 500 mm.
- C) A circular beam, 50 mm in diameter, is welded to a support by means of a fillet weld as shown in Fig. Determine the size of the weld, if the permissible shear stress in the weld is limited to 100 N/mm^2 . 6



Q6. Solve Any Two of the following.

- A) Explain in detail the design procedure for Flexible Coupling. 6
- B) A helical compression spring, made of circular wire, is subjected to an axial force, which varies from 2.5 kN to 3.5 kN. Over this range of force, the deflection of the spring should be approximately 5 mm. The spring index can be taken as 5. The spring has square and ground ends. The spring is made of patented and cold-drawn steel wire with ultimate tensile strength of 1050 N/mm^2 and modulus of rigidity of 81370 N/mm^2 . The permissible shear stress for the spring wire should be taken as 50% of the ultimate tensile strength. Design the spring and calculate 6
- wire diameter;
 - mean coil diameter;
 - number of active coils;
 - total number of coils;
 - solid length of the spring;
 - free length of the spring;
- C) A shaft, 40 mm in diameter, is transmitting 35 kW power at 300 rpm by means of Kennedy keys of $10 \times 10 \text{ mm}$ cross-section. The keys are made of steel 45C8 ($S_{yt} = S_{yc} = 380 \text{ N/mm}^2$) and the factor of safety is 3. Determine the required length of the keys. 6

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